Gating AI Deployment on Environmental Evidence Quality

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Abstract

Carbon-aware governance proposals gate AI deployment on impact magnitude: a carbon budget, a readiness level, or a risk rating. None of them asks whether the carbon number itself can be trusted. We present FairMind-E, an environmental governance module deployed inside an open AI-governance platform, that makes measurement provenance a first-class release criterion. Each assessment maps its measurement source to one of five provenance classes, which cap a numeric evidence-confidence score (vendor-reported figures cap at 0.60; undisclosed provenance forces 0.0); confidence crossed with an impact tier in a 3×3 decision matrix yields *go*, *conditional_go*, or *no_go*, and the recommendation blocks the platform's deployment-approval workflow. Offsets and renewable certificates never improve confidence, tier, or recommendation. On a 14-case hand-labeled fixture set the full mechanism reproduces all expected decisions, while ablated gates that see only carbon magnitude agree with at most half of them — a consistency check, not a field result, which is why we also present the design of two studies, instrumented and ready to run with pre-registration planned: a measurement study in which identical workloads measured three ways (power tracker, cloud billing, analytic FLOP estimate) receive different confidence scores and can flip the gate at identical true impact, and a two-arm expert panel testing whether the matrix changes reviewer coverage and agreement. We seek workshop feedback on both study designs before data collection completes.

1 The gap: a measurement layer with no trust layer

Regulators and standards bodies are building the measurement layer for AI's environmental footprint: quantification-methods inventories (CEN/CLC/TR 18145 [16]), sustainability overviews (ISO/IEC TR 20226 [17]), measurement frameworks in progress (IEEE P7100), and a G7-commissioned inventory that catalogues thirteen measurement, monitoring, and reporting initiatives [18]. The EU AI Act anticipates harmonised standards for "reporting and documentation processes to improve AI systems' resource performance" (Article 40), and GPAI technical documentation must state known or estimated energy consumption (Annex XI) [14]. Yet none of these instruments — and none of the thirteen catalogued initiatives — defines how the *quality* of an environmental claim should condition a deployment decision.

Recent governance research gates on magnitude. Carbon-Aware Governance Gates embeds carbon budgets in governance layers, with provenance recorded only in audit logs [1]; G-TRACE gates on sustainability readiness levels, arguing that "launch blocking without carbon sign-off creates financial-grade controls" [2]; risk-management integrations treat AI carbon as a risk dimension in banking frameworks [3]. Meanwhile, measurement research has established that reliability varies sharply with disclosure: metadata-completeness tiers across 5,227 public models span roughly $1.2\times$ to $2\times$ estimation error [4], and spend-based fallbacks in tiered Scope 3 accounting overestimate by 10–40$\times$ [5]. Adjacent work has begun to act on estimate trust — conformal calibration inside a carbon budget gate for agentic AI [6], confidence-gated filtering of industrial emissions predictions [7] — and evidence-driven release gates are established for functional quality [8]. The two literatures have not been connected: a system can sit comfortably under its carbon budget while its carbon number is untrustworthy, and every published gate would pass it.

FairMind-E connects them. Our claim is narrow and, to our knowledge, unheld: environmental *evidence quality* — not only impact magnitude — should gate AI deployment, operationalized as a provenance-derived confidence score crossed with an impact tier. The intellectual ancestry is the data-quality tradition in life-cycle assessment (pedigree matrices and data quality indicators [10]) and the NUSAP uncertainty pedigree [11]; recent work recommends DQI-style disclosure for AI-assisted product footprinting but proposes no score, no thresholds, and no gate [9]. We bring that tradition, inverted, into AI release governance: they assess AI that produces carbon numbers; we score carbon numbers to gate the AI.

Impact \ Confidence	measured	estimated	unknown
low	go	go	conditional_go
medium	conditional_go	conditional_go	no_go
high	conditional_go	no_go	no_go

Table 1: The impact × evidence-confidence decision matrix. Confidence exactly 0.0 (undisclosed provenance) returns *no_go* for every tier, before matrix lookup.

2 Mechanism

Provenance classes and confidence. Every assessment declares its measurement source, normalized to one of five provenance classes with bounded confidence ranges: *measured* (hardware telemetry; 0.85–1.00), *tool_estimated* (software power trackers; 0.60–0.85), *vendor_reported* (cloud-provider figures; 0.40–0.60), *manual* (analytic FLOP estimates; 0.20–0.40), and *unknown* (0.0). A caller-supplied score is honored but never allowed to exceed its provenance cap: a claimed 0.95 on vendor-reported data becomes 0.60, and unknown provenance forces 0.0 regardless of any claimed score. Provenance is categorical and reported separately from quantified uncertainty (a percent figure), so a precise-looking number cannot launder a weak source.

Impact tiers. Metrics map to low, medium, or high impact per lifecycle phase (training, fine-tuning, inference, LLM inference), using location-based carbon for tiering so market instruments cannot lower a gate outcome. The thresholds are explicitly provisional and are recalibrated from the measurement study (Section 4); an intensity-versus-baseline ratio can raise a tier at any confidence but lower it only with measured-grade confidence.

The decision matrix. Confidence bands (measured ≥ 0.70; estimated 0.35–0.69; unknown < 0.35) cross with impact tiers (Table 1). Confidence 0.0 short-circuits to *no_go* before matrix lookup: an undisclosed claim never passes. A *conditional_go* cannot be approved until it carries a dated, owned mitigation or an explicit exception (owner, expiry, rationale) — the exception path makes reviewer overrides auditable instead of informal. Six controls (ENV-1..ENV-6: boundary, energy accounting, dual location/market carbon, per-functional-unit efficiency, evidence confidence with disclosed uncertainty, mitigation/exception) define coverage, and disclosed offsets are recorded but never subtracted from emissions and never improve any gate output.

3 Implementation and consistency evaluation

FairMind-E ships inside FairMind, an open AI-governance platform, not as a standalone proposal. The decision engine is a framework-agnostic Python package (57 unit tests covering every matrix cell, band boundary, provenance cap, and the undisclosed-evidence short-circuit). Assessments persist as versioned governance evidence; a release gate hooks the platform's existing deployment-approval workflow and returns HTTP 409 with the blocking reason when the latest assessment is *no_go* or an unmitigated *conditional_go* (15 integration tests). Reviewers get a UI tab with the matrix position, control coverage, mitigation tracker, and version trail; disclosures export in an ESRS E1-shaped format [15]. The mechanism is therefore machine-checkable end to end: given the same assessment payload, the gate decision is deterministic and reproducible.

Fixture consistency check. On 14 hand-labeled cases spanning all matrix cells, mitigation states, exception paths, and offset disclosures, the full mechanism reproduces 14/14 expected labels. Ablations degrade sharply (Table 2): a carbon-only gate (magnitude thresholds, no evidence dimension) agrees on 7/14; a generic sustainability score on 6/14; no environmental gate on 3/14; removing only the exception path or the mitigation-review requirement costs 1–3 cases each. We state the boundary plainly: the authors wrote both the fixtures and the engine, so this is a consistency and ablation illustration — evidence that the confidence dimension is load-bearing in the mechanism — and not an empirical validation. That is what the two studies are for.

4 Validation studies (designed and instrumented, pre-data)

Measurement study. Four workloads (classical ML training, transformer fine-tune, RAG inference, batch embeddings) × 5 iterations, each physical run measured three ways: a software power tracker (tool_estimated), operator-filled cloud billing (vendor_reported), and a documented analytic FLOP estimate (manual). The harness never fab-

Gate variant	Exact matches	Accuracy
FairMind-E (full mechanism)	14/14	100.0%
no exception path	13/14	92.9%
offset-credit gate	13/14	92.9%
no mitigation-review gate	11/14	78.6%
carbon-only gate	7/14	50.0%
generic sustainability score	6/14	42.9%
no environmental gate	3/14	21.4%

Table 2: Hand-labeled fixture ablation (14 cases). A consistency check on author-written fixtures, not an empirical validation; reported to show the evidence-confidence dimension is load-bearing.

ricates: missing dependencies or billing entries yield `method_available=false` rows. Outputs feed method-agreement statistics (pairwise MAPE, Bland-Altman limits of agreement) that recalibrate the provisional tier thresholds and confidence ranges. The design isolates the paper's central question: the same run, at identical true impact, carries different provenance under each method — we report how often the gate decision flips on provenance alone, a quantity no magnitude-only gate can express.

Expert panel. Twelve synthetic system profiles cover all nine matrix cells (boundary cells doubled with differing mitigation states), each validated against the production engine; 15 raters (5 GRC, 5 ML engineering, 5 ESG) rate 8 profiles each under a counterbalanced assignment (10 ratings per profile, at least 3 per role). Two arms: baseline (raw measurements only) versus module (full FairMind-E output). Pre-specified hypotheses: H1, displayed confidence correlates negatively with recommendation ordinal (Spearman); H2, the module arm raises ENV control coverage against a per-profile gold checklist (Wilcoxon signed-rank on paired coverage rates); H3, the matrix improves inter-rater agreement (mean pairwise Cohen's κ and Fleiss' κ with bootstrapped CIs, target $\kappa > 0.60$). The analysis pipeline is implemented with exact statistics, environment-independent bootstrap intervals, and hand-verified test vectors, so results are computable the day collection closes.

5 Pathway to impact

The near-term impact channel is regulatory: Article 40's harmonised standards for resource-performance reporting do not yet exist, and the standards bodies' current deliverables contain no environmental item [14, 19]. A provenance-scored, platform-integrated gate is a concrete, testable proposal for what those standards could require — anticipatory infrastructure rather than commentary. The mechanism is deployed in an open platform; the fixtures, profiles, harness, and analysis code are public in the project repository; and both studies are designed for independent replication. We ask the workshop for critique of the two study designs — particularly the panel's baseline-arm construction and the defensibility of provenance caps — while changes are still cheap.

Limitations. The fixture evaluation is author-constructed and non-empirical by design; the measurement study is single-site in its first run; the panel targets 15 raters, sized for feasibility rather than power on small effects. Impact-tier thresholds remain provisional until recalibrated.

References

[1] Carbon-Aware Governance Gates (CAGG): embedding carbon budgets, energy provenance, and sustainability-aware validation in governance layers. arXiv:2602.19718v2, 2026.

[2] G-TRACE: an AI Sustainability Pyramid for carbon-aware governance readiness. arXiv:2511.04776v3, 2026.

[3] N. Tkachenko. Sustainable integration of frontier AI in banking: carbon-aware risk management frameworks. *Journal of Financial Compliance*, 2025. (Preprint: arXiv:2410.01818.)

[4] Hugging Carbon: estimation reliability tiers for the carbon footprint of 5,227 public models. arXiv:2605.01549v2, 2026.

[5] Four-tier Scope 3 methodology for AI inference under CSRD. arXiv:2606.10660, 2026.

[6] Green SARC: predictive cost and carbon governance for agentic AI with split-conformal calibration. arXiv:2606.15954, 2026.

[7] Trust-aware predictive emissions monitoring: confidence-gated filtering of industrial emissions estimates. arXiv:2606.06156, 2026.

[8] Automated self-testing as a quality gate: evidence-driven release decisions. arXiv:2603.15676, 2026.

[9] Criteria for credible AI-assisted carbon footprinting systems. arXiv:2509.00240, 2025.

[10] B. P. Weidema and M. S. Wesnæs. Data quality management for life cycle inventories — an example of using data quality indicators. *Journal of Cleaner Production*, 4(3–4):167–174, 1996.

[11] S. O. Funtowicz and J. R. Ravetz. *Uncertainty and Quality in Science for Policy*. Kluwer, 1990.

[12] Toward sustainable generative AI: a scoping review of carbon footprint methods and tools. *IEEE Access*, 14, 2026. doi:10.1109/ACCESS.2026.3659894.

[13] A. S. Luccioni, S. Viguier, and A.-L. Ligozat. Evaluating the social and environmental impact of generative AI. arXiv:2306.05949, 2023.

[14] Regulation (EU) 2024/1689 (AI Act), Article 40 and Annex XI §1(2d).

[15] Directive (EU) 2022/2464 (CSRD) and ESRS E1 (climate change), 2023.

[16] CEN/CLC/TR 18145:2025. Environmentally sustainable artificial intelligence: quantification methods inventory. 2025.

[17] ISO/IEC TR 20226:2025. Artificial intelligence — environmental sustainability aspects. 2025.

[18] G7 French Presidency. Overview of initiatives for the measurement and monitoring of energy and resource use of AI models. May 2026.

[19] Regulating the environmental footprint of AI: the regulatory landscape. arXiv:2603.00068, 2026 (FAccT).